05 FUNCTIONS • THEORY

03 recursion and stack frames

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

03 — Recursion and Call Stack Frames

Every recursive function call pushes a new activation record (stack frame) onto the call stack containing its own local variables and return address. Every recursive function **must** have a **base case** that terminates recursion; otherwise, infinite recursion exhausts stack memory (Stack Overflow).